

Priya Rao

Machine Learning Research Engineer

Singapore | priya.rao@example.com

PROFESSIONAL SUMMARY

Machine learning engineer with research and industry experience in text classification and information retrieval. Experienced in Python, PyTorch, scikit-learn, model evaluation and reproducible experiments. Seeking applied NLP or machine learning engineering roles.

TECHNICAL SKILLS

Languages: Python, SQL, C++

ML / NLP: PyTorch, scikit-learn, sentence-transformers, text classification, embeddings

Engineering: FastAPI, PostgreSQL, Docker, Git, Linux

Evaluation: Macro F1, precision, recall, NDCG@10, error analysis

WORK EXPERIENCE

Machine Learning Engineer | Example Search Technologies

Singapore | February 2025 - August 2026

- Built a semantic retrieval prototype using sentence-transformers and PostgreSQL vector search.
- Improved NDCG@10 from 0.61 to 0.69 on a fixed offline evaluation set of 500 queries.
- Exposed model inference through FastAPI and packaged the service with Docker.
- Documented dataset limitations and reviewed false-positive retrieval results with product colleagues.

Research Assistant (part-time) | Example Institute of Computing

Singapore | September 2023 - December 2024

- Conducted text classification experiments alongside full-time MSc studies.
- Evaluated a classifier on 10,000 labelled documents, achieving macro F1 of 0.84 on a held-out test split.
- Maintained reproducible experiment configurations and prepared research presentations.

EDUCATION

MSc in Artificial Intelligence | Example Institute of Computing

August 2023 - January 2025

Thesis: Robust Text Classification Under Domain Shift

BEng in Electrical Engineering | Example Engineering College

August 2018 - May 2022

RESEARCH PROJECTS

Multilingual Support Ticket Classifier | February 2024 - November 2024

- Compared TF-IDF logistic regression against a PyTorch text classifier on English and Hindi support tickets.
- Recorded per-language precision and recall to examine uneven performance.
- Used anonymised research data and documented restrictions on model reuse.

Document Retrieval Evaluation Toolkit | March 2025 - October 2025

- Created scripts to calculate NDCG@10 and recall@20 from ranked search results.
- Added repeatable baseline experiments and tests for duplicate-document handling.

PUBLICATIONS

Priya Rao and Daniel Chen. "Evaluating Domain Shift in Support Ticket Classification." Example Workshop on Applied NLP, 2024. Workshop paper.

Priya Rao. "A Reproducible Protocol for Small-Scale Retrieval Evaluation." Example Institute Technical Report TR-2025-07, 2025. Technical report, not peer reviewed.

CAREER BREAK

June 2022 - July 2023 | Family caregiving. No employment during this period.

VOLUNTEERING

Python Workshop Mentor | Example Community Learning Group

March 2024 - December 2024 | Two sessions per month

- Assisted 15 adult learners with introductory Python exercises. This was an unpaid volunteer role.

LANGUAGES

English: Fluent | Hindi: Native | Mandarin Chinese: Beginner

ADDITIONAL INFORMATION

Open to applied research and ML engineering roles in Singapore. Research assistant employment overlapped with MSc studies because it was part-time. Publication titles and organisations in this test resume are fictional.